

Statistics & Probability



Measures of center

- 1 Find the mean of 12, 15, 18, 9, 21.
 - 2 Find the median of 4, 9, 2, 11, 7, 15.
 - 3 Find the mode of 3, 7, 5, 7, 2, 7, 9.
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Measures of spread

- 4 Find the range of the data set 22, 18, 35, 9, 27.
 - 5 Find the interquartile range (IQR) of 4, 7, 9, 12, 15, 18, 21.
 - 6 A data set has a range of 40 and a minimum value of 15. Find the maximum value.
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Basic probability

- 7 A jar has 5 red, 3 blue, and 2 green marbles. Find $P(\text{green})$.
 - 8 A fair six-sided die is rolled once. Find $P(\text{even number})$.
 - 9 A spinner has 8 equal sections numbered 1 through 8. Find $P(\text{number greater than 5})$.
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Combined probability: AND / OR

- 10 A card is drawn from a standard 52-card deck. Find $P(\text{king or queen})$.
 - 11 Two independent events have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.
 - 12 A bag has 6 yellow and 4 purple marbles. A marble is drawn, replaced, then a second is drawn. Find $P(\text{both yellow})$.
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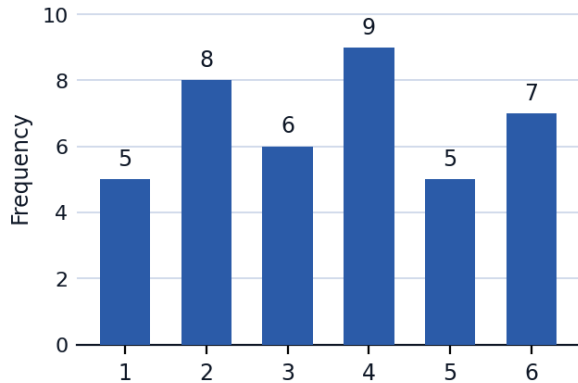
Probability without replacement

- 13 A bag has 5 red and 7 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

- 14 A committee of 2 is chosen at random, one at a time without replacement, from a group of 4 men and 6 women. Find $P(\text{both chosen are women})$.
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Data displays

- 15 The bar graph shows the results of rolling a fair die 40 times. Based on the experimental data, find the experimental probability of rolling a 4, and compare it to the theoretical probability.



Expected value

- 16 A game costs 2 dollars to play. A player wins 10 dollars with probability 0.1, and wins nothing otherwise. Find the expected net profit per play.
- 17 A spinner has 4 equal sections labeled 5, 10, 15, and 20 points. Find the expected value of one spin.
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Effects of transformations on statistics

- 18 A data set has mean 20. If every value in the set is increased by 5, find the new mean.
- 19 A data set has standard deviation 4. If every value is multiplied by 3, find the new standard deviation.
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A well-designed word problem: a survey of students

- 20** This question has three parts. A researcher surveys 200 students about whether they play a sport and whether they play a musical instrument. Of the 200 students, 120 play a sport, 80 play an instrument, and 30 do both. (a) Find how many students do neither activity. (b) Find the probability that a randomly chosen student plays a sport or an instrument (or both). (c) Given that a randomly selected student plays a sport, find the probability that they also play an instrument.